REPLY × LUISS 2026 — PROJECT 2

Classical vs Multi-Agent

Anomaly Detection on Italian Border-Control Data

Same detection logic, two architectures — measured.

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The Brief — what we were asked

EXPLICIT REQUIREMENTS

Classical pipeline:

step-by-step notebooks with IF / LOF / Z-score on a historical baseline.

Multi-agent system:

5 agents — Data, Baseline, Outlier, Risk Profiling, Report.

Compare both architectures and recommend.

IMPLICIT — WHAT IT REALLY MEANS

Numbers must converge.

If the two pipelines disagree, one is buggy. Convergence is the test, not the problem.

Differentiation lives elsewhere.

Operational UX, modularity, explainability, single-agent re-run, failure isolation.

Small data — model accordingly.

Single deliverable: main.ipynb (end-to-end runnable) + README + PPTX.

The Data — Italian airport border-control records

567
unique routes
(airport pair × month aggregated)

3
months
Dec 2023 + Jan/Feb 2024

54
engineered features
per route

2
raw CSVs
ALLARMI +
TIPOLOGIA_VIAGGIATORE

THE BUSINESS QUESTION

Is this route behaving anomalously compared with the rest of the population?

Outliers in alarm rates (Interpol, SDI, NSIS), closure rates, rejection rates, and traveller-mix profiles.

Two Architectures, Same Logic

CLASSICAL PIPELINE

Linear orchestrator.
Six step functions, each persists output to disk before the next runs.

Properties:
deterministic · audit-friendly · ~3 s end-to-end · simple to reason about.

Limit:
rigid. Any new perimeter triggers a full re-run.

MULTI-AGENT (LangGraph)

DAG with 4 conditional edges.
Five agents communicate only via shared AgentState (no direct imports).

Properties:
modular · runtime perimeter filter · LLM narratives · ~1.6 s end-to-end (no LLM).

Limit:
more code surface to maintain.

Shared by construction: preprocessing module · FeatureBuilder · MAD baseline · 5 business rules · ensemble weights.

Classical Pipeline — six sequential steps

0 Preprocess
Clean + merge
2 raw CSVs

▶

1 Features
FeatureBuilder
54 features / route

▶

2 Baseline
MAD z-scores
+ Tukey audit

▶

3 Detection
IF · LOF · Z · AE
weighted ensemble

▶

4 Post-proc
5 business rules
final_risk

▶

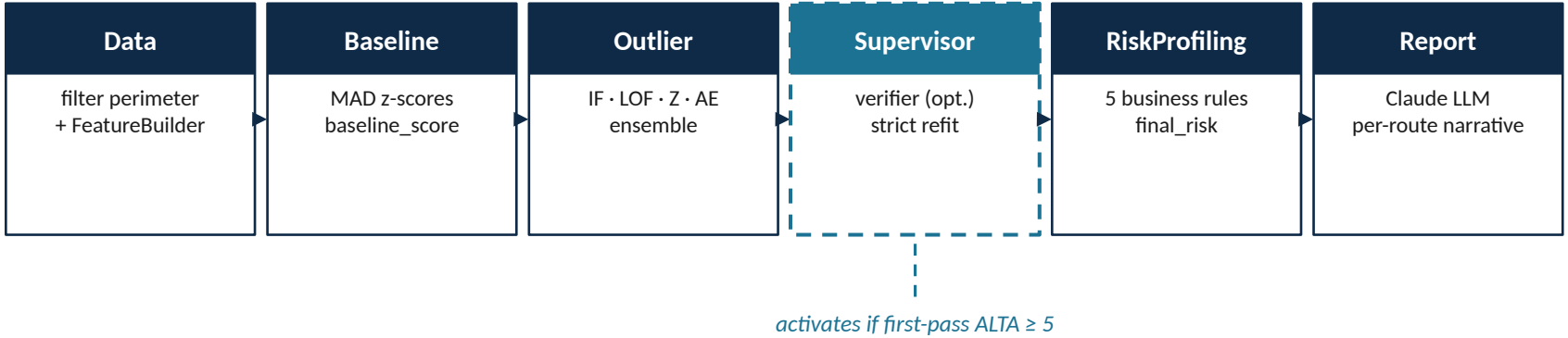
5 Evaluate
Silhouette · bootstrap
feature importance

EACH STEP PERSISTS ITS OUTPUT

data/processed/ → dataset_merged.csv → features_classical.csv → features_with_baseline.csv → anomaly_results.csv → final_report.csv → evaluation_scorecard.json

Reproducible: random_state = 42 across IF / LOF / MLPRegressor / GradientBoosting / bootstrap RNG.

Multi-Agent Topology — 5 spec agents + 1 verifier



STATE CONTRACT

AgentState (LangGraph TypedDict). Each agent reads what its predecessors wrote and writes only its own keys. ***No agent imports another.***

Five mandated agents match the spec; **Supervisor** is a non-mandatory verifier — re-fits IsolationForest at contamination = 3 % on the ALTA subset and tags `alta_robusta` for routes that survive the stricter rule.

Four Conditional Edges — operative vs guard rail

Two reshape the work the pipeline does on the typical input. Two are safety valves for edge cases.

EDGE	DECISION	KIND	WHEN IT FIRES
after_outlier	ALTA ≥ 5 → Supervisor else → Risk	OPERATIVE	Always: 17 ALTA on 2024 → verifier; 0 ALTA on Marocco → skip
after_risk	ALTA/MEDIA present → Report else → end	OPERATIVE	Skips LLM cost on quiet perimeters
after_baseline	n_features < 5 OR std < 0.01 → end else → Outlier	GUARD RAIL	Triggers on degenerate slices (e.g. paese=Algeria, 2 routes)
after_supervisor	downgrade > 50% AND iter < 2 → Outlier (cycle) else → Risk	GUARD RAIL	Inactive on standard data (29% downgrade); cycle capped at 2 iter

Detection Ensemble — four complementary signals

0.35
ensemble weight

IsolationForest

tree-based · n_estimators 200 · contamination 0.10

0.30
ensemble weight

Local Outlier Factor

density-based · k = 20 · catches local anomalies

0.15
ensemble weight

MAD z-score

median ± 1.4826·MAD · std fallback when MAD = 0

0.20
ensemble weight

Autoencoder MLP

11→8→4→8→11 · semi-supervised on normal mass

Data-driven thresholds: p97 of ensemble_score → ALTA, p90 → MEDIA. Floors at 0.30 / 0.20 to avoid noise on quiet perimeters.
Graceful degradation: Autoencoder excluded with < 30 normal samples; remaining weights renormalised (0.41 / 0.35 / 0.18 / 0).

Business Rules — symbolic layer on top of ML

RULE	CONDITION	BUSINESS MEANING
br_high_interpol	pct_interpol ≥ 0.30	*Suspicious Interpol alarm rate*
br_high_rejection	tasso_respinti ≥ 0.25	*Pattern of active rejections*
br_low_closure	tot_allarmi_log > 3 AND tasso_chiusura < 0.10	*Many alarms, almost none closed*
br_multi_source	pct_interpol > 0 AND pct_sdi > 0	*Multi-channel signal — stronger*
br_high_alarm_rate	tasso_allarme_medio ≥ 0.50	*Average alarm rate above threshold*

CONFIDENCE = 0.60 · anomaly_score + 0.40 · br_score → final_risk ∈ { CRITICO, ALTO, MEDIO, BASSO }

ML score and rule score combined; 60/40 ratio chosen to keep ML dominant but allow business overrides on borderline cases.

Numerical Convergence — both pipelines on 567 routes

98.24 %

label agreement

557 / 567 routes share the same ALTA / MEDIA / NORMALE label

COMPONENT	PEARSON R
IsolationForest	1.0000
Local Outlier Factor	1.0000
Z-score (shared MAD)	1.0000
Autoencoder MLP	0.9663
ensemble_score	0.9965
Spearman rank ρ	0.9980

Identical risk distribution: 17 ALTA · 40 MEDIA · 510 NORMALE. **Top-10 overlap 9/10 · Top-50 overlap 46/50.**

Convergence is the certificate, not the problem

If two implementations of the same logic disagree numerically, one of them is buggy.

The 98.24 % agreement is a structural cross-validation between architectures.

THE 1.8 % RESIDUAL — WHAT IT IS

10 routes, all sitting at the MEDIA / NORMALE boundary.

Driven entirely by the Autoencoder.

MLPRegressor with random_state = 42 is non-deterministic across pandas row-orderings, which differ between pipeline runs.

The 10 routes lie within ±0.005 of the p90 threshold — any AE noise reorders them.

WHAT IT IS NOT

Not a baseline mismatch.

Both pipelines now share the same MAD baseline by construction — Z-component r = 1.0000.

Not a feature drift.

FeatureBuilder and preprocessing are imported from the same modules; no copy-paste.

Not a thresholding artefact.

Both apply identical p97/p90 cuts on the ensemble.

Statistical Robustness — three layers beyond the headline

BOOTSTRAP CI

1000

iterations × 80 % subsample, no replacement.
Confidence interval on the 98.24 % agreement.
Implicit regression test: drift detection at zero cost.

THRESHOLD SENSITIVITY

±20%

Perturbation of the five business-rule thresholds by ±20 %.
Measures fragility of final_risk to operator-tuned cut-offs.
Locates the rules whose calibration matters most.

TREND SLOPE

Δ/m

Linear slope of tasso_allarme_medio per route, over the 3-month window of the dataset.
A pragmatic substitute for STL on a short panel where most routes appear in 1–2 of the 3 months.

All three layers run inside main.ipynb, on the same data the headline numbers come from. Reproducible from a single command.

Why the multi-agent layer earns its keep

The numbers are the same. The advantage is operational, measurable, and demoed live in Streamlit.

1 Runtime perimeter filter

{anno, paese, aeroporto, zona} at runtime → only the matching subset traverses the graph in ~0.5 s. Classical re-runs preprocessing + features + baseline + ML for every new perimeter.

2 Granular re-run

Change a business-rule threshold? Only RiskProfilingAgent re-runs (~10 ms). Classical replays the four ML stages above it.

3 Per-route LLM narratives

ReportAgent (Claude) explains each ALTA/MEDIA route citing both z-drivers and triggered business rules. Classical produces JSON only — non-actionable for an operator.

4 Failure isolation

If the LLM API is down, ReportAgent is skipped and all numerical output is preserved. Each agent fails in place; no cascading silent corruption.

Pick the architecture by use case, not by reflex

CHOOSE CLASSICAL WHEN

- **Scheduled batch jobs**
  - Run nightly on the full dataset, no perimeter changes.
- **CI / regression checks**
  - Deterministic invariants on a fixed score.
- **Audit trails as the deliverable**
  - Each step persists; reviewers inspect intermediate CSVs.
- **Constrained operational footprint**
  - Smaller surface, fewer moving parts to maintain.

CHOOSE MULTI-AGENT WHEN

- **Interactive dashboards**
  - Streamlit user filters perimeter, expects sub-second feedback.
- **LLM-augmented reporting**
  - Per-route narratives an operator can act on.
- **Frequent rule changes**
  - Threshold tuning without rerunning the ML stack.
- **Failure isolation matters**
  - External APIs (LLM) can fail without taking the pipeline down.

What's weak — and what we'd do next

P1

Formal test coverage

One pytest file. Cross-pipeline agreement is our regression test in disguise. Real test suite is the first investment for a v2.

P1

Autoencoder determinism

MLPRegressor non-determinism survives random_state=42 because pandas row-ordering varies. A sort_values('ROTTA') before fit closes the gap.

P2

Supervisor cycle is dormant

downgrade_rate threshold at 50 %; on real data the cycle never fires (29 %). Sell as safety valve, not active feature — or recalibrate.

P2

Notebook size

1.84 MB · 219 cells. Bootstrap + sensitivity + trend belong in an appendix; the main flow should be 70 % shorter.

P3

Per-agent latency telemetry

elapsed_s is logged but not surfaced. Production telemetry (p50/p95) per agent would close the observability gap.

P3

Hyperparameter tuning

Ensemble weights and contamination set by hand. A grid-search with bootstrap-CV would justify them quantitatively.

P1 = ship-blocker if we kept building · P2 = obvious next iteration · P3 = polish

RECAP

Two architectures, one detection system.

The numbers converge by construction; the architecture choice is operational.

DETECTION

98.24 % label agreement.
Perfect IF / LOF / Z correlation.
Identical 17 / 40 / 510 distribution.

ARCHITECTURE

5 agents matching the spec + 1 verifier.
Four conditional edges, two operative.
LangGraph DAG, ~1.6 s end-to-end.

RECOMMENDATION

Classical for batch jobs and audit trails.
Multi-agent for interactive dashboards, LLM narratives, and rapid rule iteration.

Thank you — questions welcome.

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